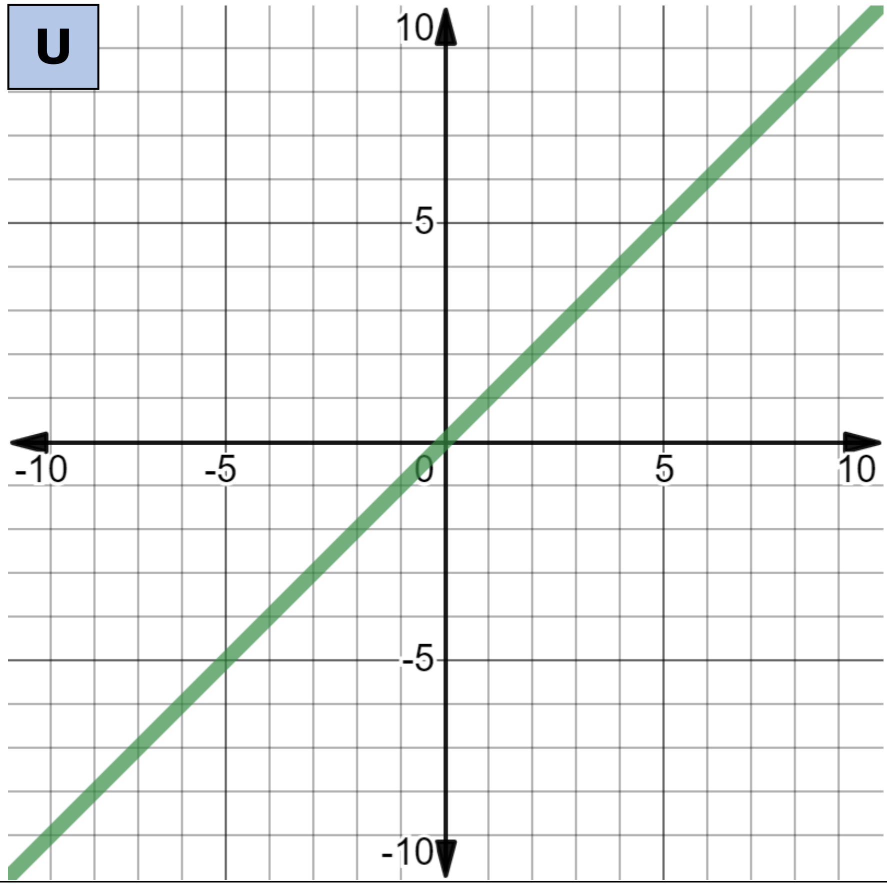
13.4.2 Activity

V

$$f(x) = x$$

U



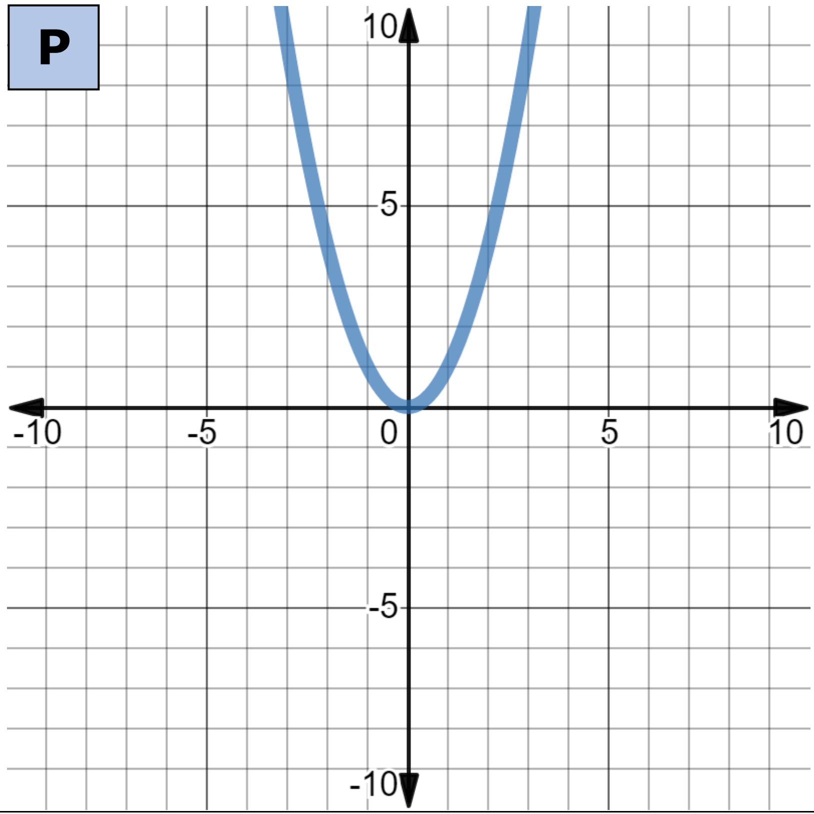
N

Key Feature Name	Key Feature
domain	$\{x: x \in \mathbb{R}\}$
range	$\{y: y \in \mathbb{R}\}$
x-intercept(s)	$(0, 0)$
y-intercept(s)	$(0, 0)$
slope	1
vertex	none
axis of symmetry	none
intervals where the function is increasing and decreasing	increasing: $\{x: x \in \mathbb{R}\}$
intervals where the function is positive or negative	negative: $\{x: x < 0\}$ positive: $\{x: x > 0\}$
end behavior	as $x \to \infty$, $y \to \infty$ as $x \to -\infty$, $y \to -\infty$

C

$$f(x) = x^2$$

P



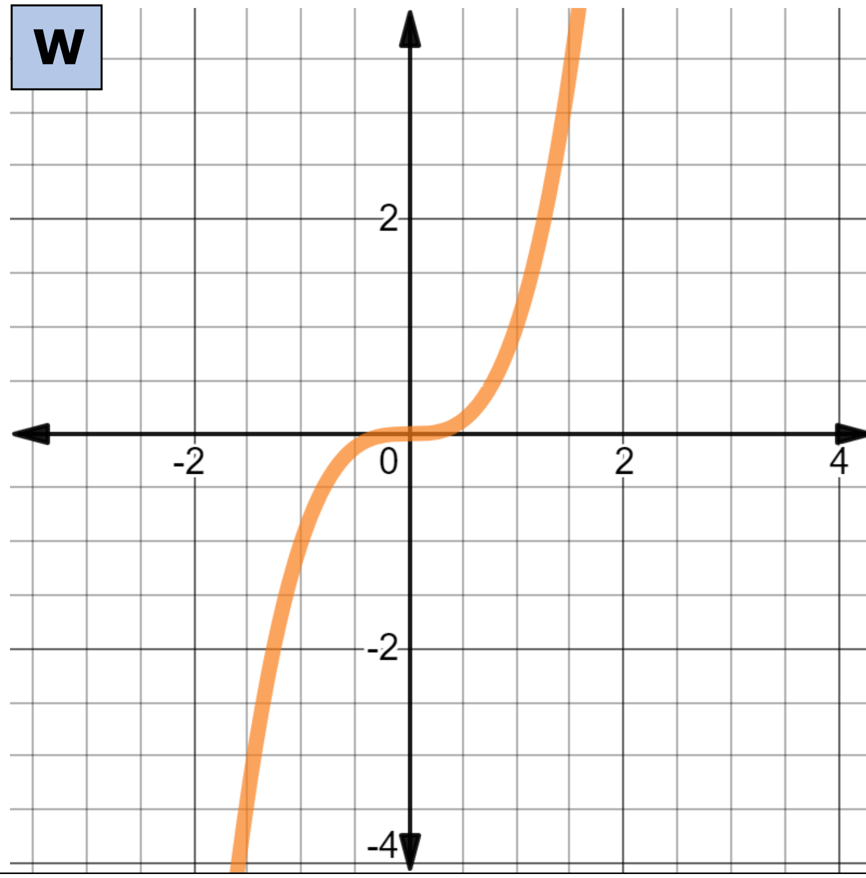
S

Key Feature Name	Key Feature
domain	$\{x : x \in \mathbb{R}\}$
range	$\{y : y \geq 0\}$
x-intercept(s)	$(0, 0)$
y-intercept(s)	$(0, 0)$
slope	none
vertex	$(0, 0)$
axis of symmetry	$x = 0$
intervals where the function is increasing and decreasing	increasing: $\{x : x > 0\}$ decreasing: $\{x : x < 0\}$
intervals where the function is positive or negative	positive: $\{x : x \neq 0\}$
end behavior	as $x \to \infty$, $y \to \infty$ as $x \to -\infty$, $y \to \infty$

D

$$f(x) = x^3$$

W



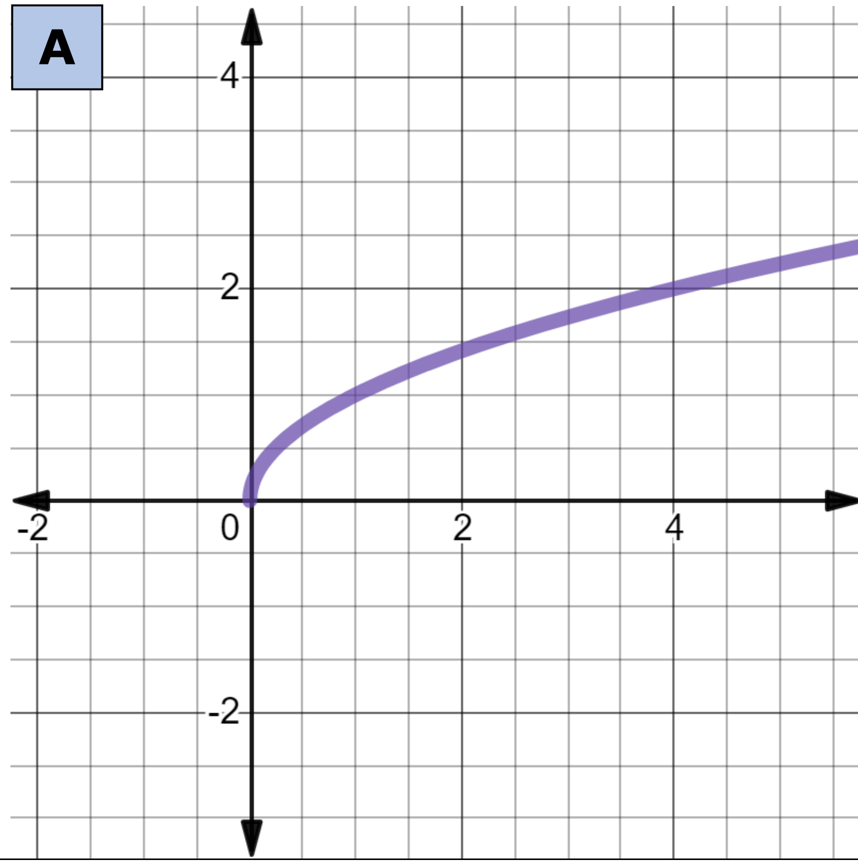
K

Key Feature Name	Key Feature
domain	$\{x: x \in \mathbb{R}\}$
range	$\{y: y \in \mathbb{R}\}$
x-intercept(s)	$(0, 0)$
y-intercept(s)	$(0, 0)$
slope	none
vertex	none
axis of symmetry	none
intervals where the function is increasing and decreasing	increasing: $\{x: x \in \mathbb{R}\}$
intervals where the function is positive or negative	negative: $\{x: x < 0\}$ positive: $\{x: x > 0\}$
end behavior	as $x \to \infty$, $y \to \infty$ as $x \to -\infty$, $y \to -\infty$

X

$$f(x) = \sqrt{x}$$

A



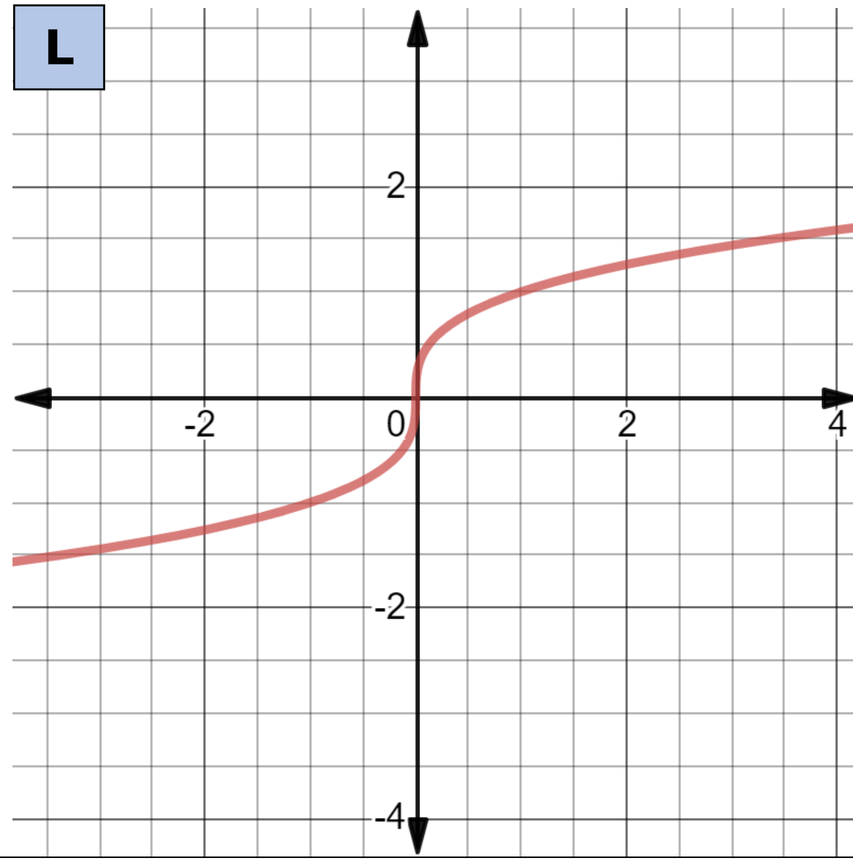
M

Key Feature Name	Key Feature
domain	$\{x: x \geq 0\}$
range	$\{y: y \geq 0\}$
x-intercept(s)	$(0, 0)$
y-intercept(s)	$(0, 0)$
slope	none
vertex	none
axis of symmetry	none
intervals where the function is increasing and decreasing	increasing: $\{x: x \geq 0\}$
intervals where the function is positive or negative	positive: $\{x: x > 0\}$
end behavior	as $x \to \infty$, $y \to \infty$

F

$$f(x) = \sqrt[3]{x}$$

L



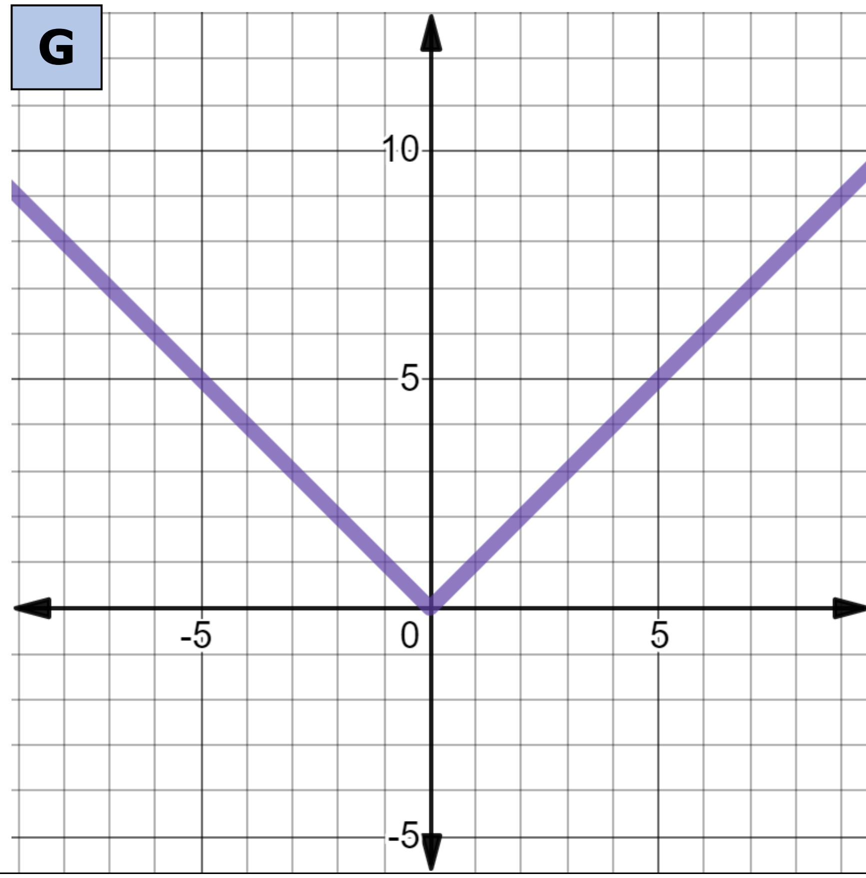
T

Key Feature Name	Key Feature
domain	$\{x: x \in \mathbb{R}\}$
range	$\{y: y \in \mathbb{R}\}$
x-intercept(s)	$(0, 0)$
y-intercept(s)	$(0, 0)$
slope	none
vertex	none
axis of symmetry	none
intervals where the function is increasing and decreasing	increasing: $\{x: x \in \mathbb{R}\}$
intervals where the function is positive or negative	negative: $\{x: x < 0\}$ positive: $\{x: x > 0\}$
end behavior	as $x \to \infty$, $y \to \infty$ as $x \to -\infty$, $y \to -\infty$

B

$$f(x) = |x|$$

G



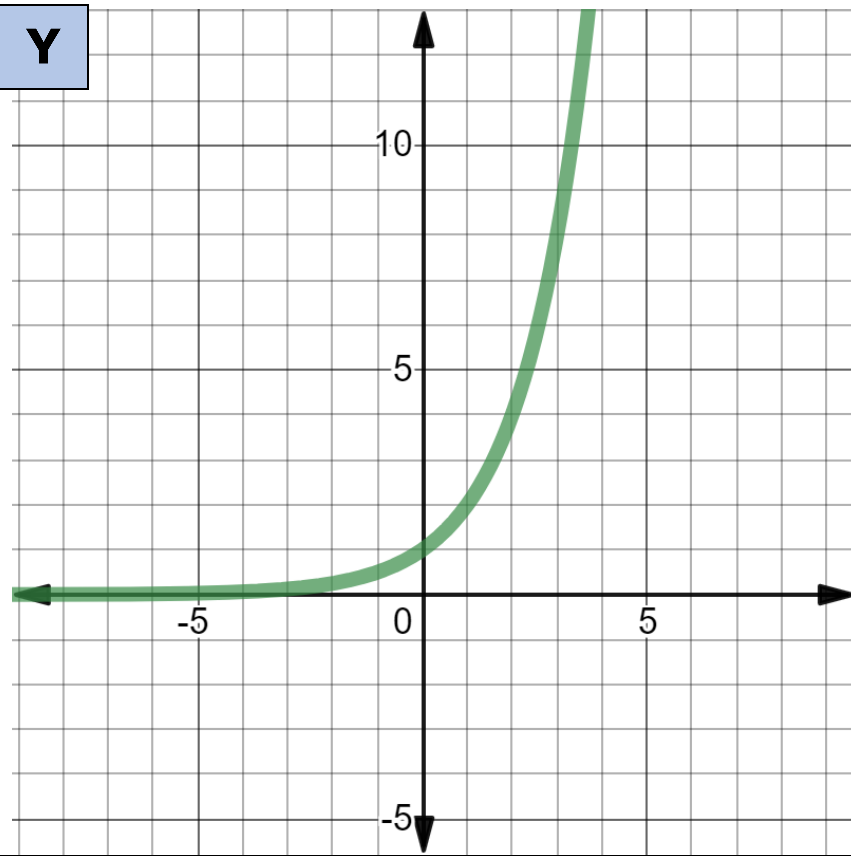
Z

Key Feature Name	Key Feature
domain	$\{x: x \in \mathbb{R}\}$
range	$\{y: y \geq 0\}$
x-intercept(s)	$(0, 0)$
y-intercept(s)	$(0, 0)$
slope	none
vertex	$(0, 0)$
axis of symmetry	$x = 0$
intervals where the function is increasing and decreasing	increasing: $\{x: x > 0\}$ decreasing: $\{x: x < 0\}$
intervals where the function is positive or negative	positive: $\{x: x \neq 0\}$
end behavior	as $x \to \infty$, $y \to \infty$ as $x \to -\infty$, $y \to \infty$

R

$$f(x) = 2^x$$

Y



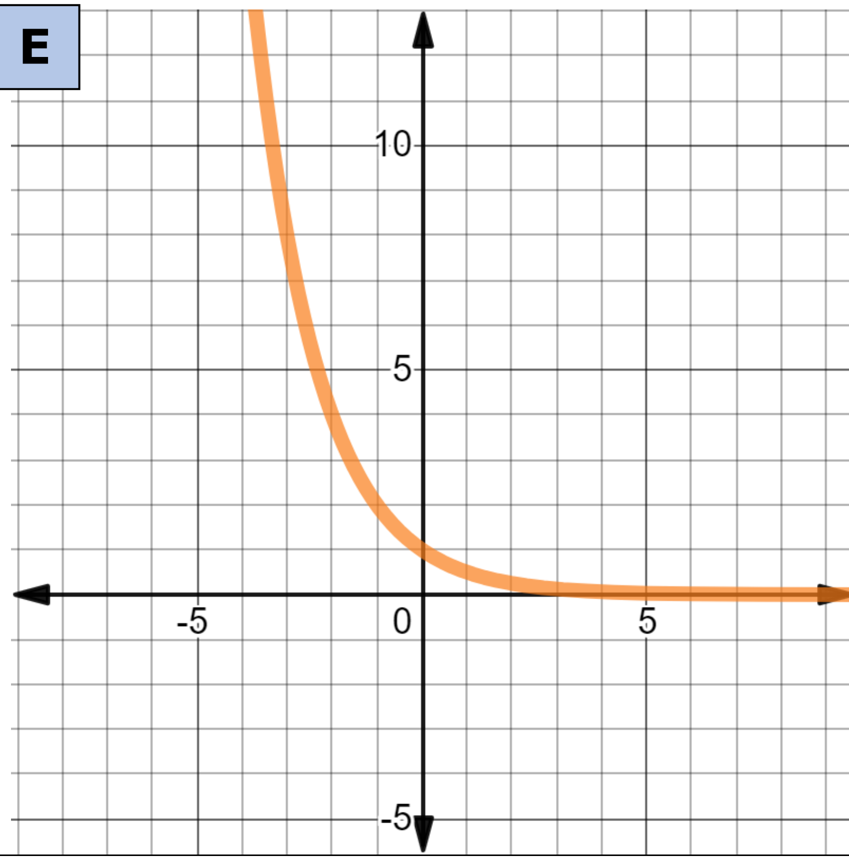
H

Key Feature Name	Key Feature
domain	$\{x : x \in \mathbb{R}\}$
range	$\{y : y > 0\}$
x-intercept(s)	none
y-intercept(s)	$(0,1)$
slope	none
vertex	none
axis of symmetry	none
intervals where the function is increasing and decreasing	increasing: $\{x : \in \mathbb{R}\}$
intervals where the function is positive or negative	positive: $\{x : x \in \mathbb{R}\}$
end behavior	as $x \to \infty$, $y \to \infty$ as $x \to -\infty$, $y \to 0$

Q

$$f(x) = \left(\frac{1}{2}\right)^x$$

E



O

Key Feature Name	Key Feature
domain	$\{x: x \in \mathbb{R}\}$
range	$\{y: y > 0\}$
x-intercept(s)	none
y-intercept(s)	$(0,1)$
slope	none
vertex	none
axis of symmetry	none
intervals where the function is increasing and decreasing	decreasing: $\{x: \in \mathbb{R}\}$
intervals where the function is positive or negative	positive: $\{x: x \in \mathbb{R}\}$
end behavior	as $x \to \infty$, $y \to 0$ as $x \to -\infty$, $y \to \infty$